Lorin Urbantat

Computer Science Student @ ETH Zurich · Software Engineer @ ETH-CES

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Education

ETH Zurich, MSc Computer Science 2025 - 2027

- Expected Graduation: September 2027
- Major: Data Management Systems
- **Information Security Lab (5.75)**, Computational Semantics for Natural Language Processing (5.75), Advanced Formal Language Theory (5.5)

ETH Zurich, BSc Computer Science 2021 - 2024

- Graduated August 2024, GPA 5.04 (6.0 – highest)
- **Parallel Programming (5.75)**, Numerical Methods (5.75), Analysis (5.5), Visual Computing (5.75), **Applied Computer Architecture (6)**

Waldorfschule Mülheim an der Ruhr, Abitur 2007 - 2020

GPA 1.3 (1.0 – highest)

Deutsche Schüler Akademie 2019

Summer school for gifted students

Work Experience

Information Security Lab Teaching Assistant, ETH Zurich 2026 - Present

Test lab exercises and host exercise sessions

Software Engineer, ETH Chair of Education Systems 2025 - Present

Sole engineer responsible for end-to-end development of a research-focused RAG chatbot using Next.js, Convex and Microsoft Foundry, serving 1,000 users

Software Engineer, ETH Game Technology Center 2023 - 2025

- Morph Tales Game, Unity 3D
- Developed Unity features for Morph Tales and built a game analytics pipeline using Firebase, BigQuery and Looker Studio.

Coach, mint & pepper 2025

STEM and AI tutor for high school students

Calculus Teaching Assistant and Exam-Preparation Course, ETH Zurich 2023

Host exercise classes, 15 - 30 students

Tutor, Waldorfschule Mülheim an der Ruhr 2017 - 2019

Tutorial classes (10 - 25 participants) in Math and Computer Science

Personal Projects & Relevant Experience

Unikraft on Raspberry Pi 5, ETH Semester Project 2026

- Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers
- Developed a cache-safe kernel reload mechanism using OpenOCD and GDB, enabling repeated deployment without power-cycling.

Flateroo, Flat sharing platform 2024 - 2025

- React Native application with real-time chat, video streaming infrastructure
- Node backend, video transcoding, HLS streaming, PostgreSQL, fly.io

OpenDemocracy, RAG Powered Voting Advice application for elections 2024

SCION Control Plane Algorithms Test Environment, Bachelor Thesis 2024

Built a Docker-based environment and traffic generator to evaluate SCION control-plane algorithms under configurable topologies and changing link conditions.

TLDR, Lecture Summarizer (VIScon Hackathon) 2023

Lecture summarizer using OpenAI Whisper to create searchable lecture summaries from videos

Skills & Interests

German (native), English (fluent), French (DELF B1) — Languages

AArch64 Assembly, C, C++, Kernel Development, TypeScript, React, Python, Java, Git, Linux, Containerization, Unity 3D, React Native, Node.js, SQL, RAG, Machine Learning, PCA — Technologies

Biking, Sports, Cooking, Dancing, Singing — Interests